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Chapter 1

Introduction

Nowadays we already have smart surveillance systems that detects the people and abnormal events, but these systems can become intelligent if it has the ability of doing the analysis and the give insights on various events happening on realtime as well as on the historic data. This can be achieved by integrating the Computer Vision and Large Language Models.

1.1 Problem Definition

This fast growing world the surveillance is generating the tremendous ammount of data. Doing analysis on these video footages is time consuming and tedious job. Also it is inefficient way of dealing with the analyzing and getting insights from it. Hence integrating the LLMs with the CV models of event detection can be very useful in solving this problem.

This solution has many realtime implementations such as crowd monitoring, Resource management at superstores, Threat management in warehouses, Monitoring parking, manage parking lots, crowd density management, fire safety monitoring and lot more. The insights generated from the LLMs can be very useful in analysis of the historic events as well along with the timesteps.

1.2 Objectives

- Develop intelligent Event Detection
- Implement CV-LLM integration
- Improve efficiency of surveillance analysis
- Provide historical event analysis

1.3 Scope

- Real time Deployment on the edge device for intelligent surveillance
- Forecasting the resource management in superstores or warehouses
- Crowd management at public places and events.

Chapter 2

Literature Survey

Over the past ten years, advances in computer vision, deep learning, and natural language processing have led to a rapid evolution in intelligent video surveillance. Numerous studies show that contemporary CV models can automate anomaly detection, event detection, and high-level video comprehension. Large Language Models (LLMs) have demonstrated impressive reasoning abilities over structured or semi-structured data in more recent times, allowing for insight generation and intelligent forensic querying. Important research on (i) CV-based event detection, (ii) surveillance video comprehension, and (iii) LLM-based structured data reasoning is reviewed in this chapter. Together, these studies support the viability and rationale of developing a surveillance intelligence system that integrates CV and LLM.

2.1 Related Works

Large Language Models for Video Surveillance Applications (2025)[1]

Vision-Language Models (VLMs) were applied to surveillance footage by De Silva et al[1]. The authors show that models like Gemini Pro Vision can produce temporally consistent summaries of complex events on datasets like MSR-VTT and campus CCTV recordings. Although the model size restricts edge deployment, their work demonstrates how LLMs can interpret multi-step event sequences. Our system's primary finding is that LLMs can accurately contextualize and summarize surveillance events, which supports the way we want to design our LLM based post surveillance analysis[1].

YOLOv7: State-of-the-Art Real-time Object Detection (2022)[2]

YOLOv7, a high-speed and high-accuracy detector that outperforms many contemporary MS COCO architectures, is proposed by Wang et al[2]. Their work presents novel model scaling methods and training strategies that can be applied in real-time CCTV scenarios. YOLOv7 validates the selection of YOLO-based backbones in our system by demonstrating that event detection on live surveillance feeds can satisfy both accuracy and latency requirements.

YOLOv8 Architecture and Feature Enhancements (2024)[3]

A comprehensive analysis of YOLOv8, describes architectural improvements such as feature pyramid networks, CSPNet and anchor-free detection [3]. YOLOv8 is an excellent choice for surveillance use cases due to its excellent performance for object, vehicle, and person detection. The anchor-free approach reduces computation while improving small-object detection for situations such as abandoned objects or crowd analysis.

FLAME: Fire Detection Using DNN + Motion Modeling (2025)[4]

The paper proposes fire detection YOLOv* based model with motion analysis and background modeling[4]. The three step method used by them helps in reducing false positives which is a major challenge in a critical event detection like fire. Paper shows domain specific events like fire, flash and smoke. This study and model can help achieve our motive of capturing critical event like fire into detection logs which later can be queried by LLM.

Abnormal Event Detection Using CCTV Camera (2021)[5]

Joemon et al. propose a CNN-LSTM autoencoder that models normal behavior in surveillance scenes and flags anomalies using reconstruction error[5]. Tested on UCSD and Avenue datasets, the method effectively detects unusual crowd movements. This work supports the principle that temporal context is essential for reliable event detection in CCTV footage. It also reinforces the need for structured logs that capture event duration and temporal relationships before sending them to an LLM.

Redefining Structured Retrieval via LLMs (ChatLR, 2024)[6]

Wang et al. introduce *ChatLR*, a system enabling LLMs to translate natural-language queries into API/SQL operations[6]. The model achieves near-perfect accuracy in structured information retrieval. This paper directly influences our design: it proves that LLMs can operate on structured surveillance logs, mapping user queries to filtering, sorting, and aggregating event databases.

HyST: Hybrid Retrieval on Semi-structured Tabular Data (2025)[7]

Myung et al. propose HyST, which uses an LLM to break down user queries into structured filters and semantic preferences[7]. This hybrid retrieval approach improves performance on datasets combining metadata and free text. For our system, HyST provides evidence that LLMs can process logs consisting of numeric, categorical, and textual fields—exactly the structure of CCTV event logs.

StructuredRAG: JSON Response Formatting with LLMs (2024)[8]

Shorten et al. evaluate how well LLMs follow strict JSON output formats[8]. Their findings show that LLMs perform well on simple structured outputs but struggle on complex nested formats. This insight is important for designing our system's event-log schema and instructing the LLM to return machine-readable results for downstream dashboards.

LLM-based Event Log Analysis Survey (2025)[9]

Akhtar et al. survey multiple strategies for using LLMs to interpret large log datasets[9]. The study concludes that LLMs significantly reduce manual auditing efforts and can detect hidden correlations in event logs. This validates the role of LLMs in forensic surveillance analysis, especially for temporal and relational querying across many cameras.

Event Detection in Surveillance Videos: A Review (2022)[10]

Karbalaie et al. provide a comprehensive review of video-event detection pipelines, including 3D CNNs, multi-stage detectors, and temporal analysis networks[10]. They highlight the complexity of surveillance events such as abandoned objects, intrusion, aggression, or anomalous motion. This reinforces the importance of modular CV event detectors in

our proposed system and justifies the use of structured logs for downstream reasoning.

2.2 Summary of Literature

Table 2.1 provides a consolidated overview of the surveyed papers, datasets used, core techniques, and relevance to our problem.

Sr.	Title	Dataset	Model / Core Tech	Take Away	Ref.
1	Large Language Models for Video Surveillance Applications	MSR-VTT, CCTV	Gemini Pro Vision (VLM)	Shows VLMs can generate reliable CCTV summaries; heavy for edge deployment but useful for high-level reasoning.	[1]
2	YOLOv7 Real-time Object Detection	MS COCO	YOLOv7	High-accuracy, real-time detector suitable for CCTV feeds; validates real-time event detection feasibility.	[2]
3	YOLOv8 Feature Analysis	MS COCO	YOLOv8	Strong detection performance; anchor-free design reduces computation for CCTV tasks.	[3]
4	FLAME: Fire Detection via DNN + Motion	Fire datasets, LFDN	YOLOv8 + background modeling	Hybrid models reduce false positives; demonstrates domain-specific event detection reliability.	[4]
5	Abnormal Event Detection in CCTV	UCSD, Avenue	CNN-LSTM Autoencoder	Detects anomalies via reconstruction error; supports temporal event modeling.	[5]
6	ChatLR: Structured DB Retrieval via LLMs	FinChina SQL DB	ChatGLM LLM	LLMs can map NL queries to structured operations, enabling forensic querying on event logs.	[6]
7	HyST Hybrid Retrieval	Amazon STaRK dataset	LLM + hybrid search	Shows LLMs can handle semi-structured tables similar to CCTV logs.	[7]
8	StructuredRAG JSON Formatting	JSON benchmarks	GPT, LLaMA	Evaluates LLM structured-output reliability; informs our output design.	[8]
9	LLM-based Log Analysis Survey	Various log corpora	LLMs (RAG, FT)	LLMs reduce manual log-review effort and surface hidden patterns.	[9]
10	Surveillance Event Detection Review	TRECVID SED, ActEV	3D CNNs + multi-stage detectors	Reinforces difficulty of surveillance event detection; validates modular CV-LLM pipeline.	[10]

Table 2.1: Summary of literature relevant to CV-based event detection and LLM-based structured surveillance analytics.

Chapter 3

Implementation

The implementation details of the proposed CV-LLM-based intelligent surveillance system is given in this chapter. The system integrates multiple CCTV streams, event detection models, a centralized storage architecture, and a Large Language Model (LLM) to enable automated analysis of real-time and historical surveillance data. The workflow is designed for modularity, scalability, and efficient forensic querying.

3.1 System Architecture Overview

The complete processing pipeline consists of four major modules:

1. **CCTV Camera Nodes with CV Models**
2. **Central Database and System Log Storage**
3. **LLM-based Analytics Engine**
4. **Human Query Interface and Insight Retrieval**

Figure 3.1 illustrates the end-to-end flow of data and insights.

The system follows a modular design where the perception layer (CV models) continuously generates structured event logs, while the reasoning layer (LLM) interprets, correlates, and retrieves insights based on user queries.

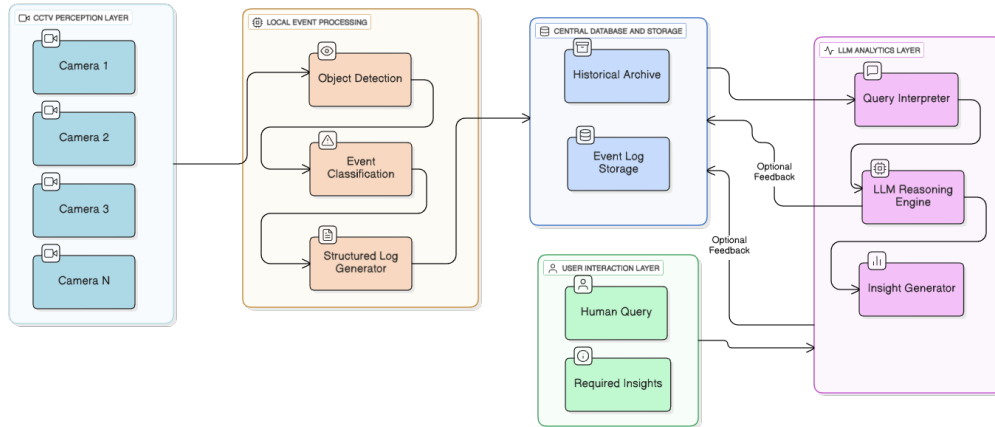


Figure 3.1: Overall System Flow Integrating CV Event Detection with LLM-based Forensic Querying

3.2 CCTV Nodes and Computer Vision Event Detection

Each CCTV stream is connected to a lightweight CV inference module capable of detecting a range of surveillance-focused events. Models such as YOLOv8, YOLOv9, or other state-of-the-art detectors are deployed depending on accuracy–latency requirements.

The CV inference module performs:

- Object detection (persons, vehicles, bags, etc.)
- Event classification (entry, exit, loitering, abandoned object, etc.)
- Tracking for temporal event interpretation (via DeepSORT or ByteTrack)
- Metadata extraction (timestamp, camera ID, confidence score)

Research indicates that several high-priority CCTV event types—such as vehicle entry/exit, abandoned objects, crowd formation, and intrusion detection—are well-supported by existing computer vision literature and datasets and therefore suitable for robust deployment[?].

Raw video data is not transferred; instead, only structured JSON logs are pushed to the central database for performance and privacy considerations.

3.3 Central Database & System Log Storage

All CV-generated structured logs are transmitted to a central storage layer responsible for:

- Aggregating event logs from all camera nodes
- Indexing logs by event type, timestamp, camera ID, and location
- Maintaining historical archives for retrospective forensic querying
- Serving as the knowledge base for the LLM analytics engine

3.3.1 Sample Logbase

here is a sample logbase :

Listing 3.1: sample Logbase

```

1  [
2    {
3      "timestamp": "2025-12-01T13:32:32",
4      "camera_id": "CAM_08",
5      "event_type": "car_entry",
6      "object_id": "660",
7      "confidence": 0.98,
8      "camera_location": "Gate_01",
9      "head_count": 0
10  },
11  {
12    "timestamp": "2025-12-06T20:06:47",
13    "camera_id": "CAM_07",
14    "event_type": "person_entry",
15    "object_id": "299",
16    "confidence": 0.8,
17    "camera_location": "Reception",
18    "head_count": 0
19  },
20  {

```

```

21     "timestamp": "2025-12-05T19:27:50",
22     "camera_id": "CAM_04",
23     "event_type": "car_exit",
24     "object_id": "64",
25     "confidence": 0.96,
26     "camera_location": "Gate_01",
27     "head_count": 0
28 },
29 {
30     "timestamp": "2025-12-01T09:30:27",
31     "camera_id": "CAM_03",
32     "event_type": "person_exit",
33     "object_id": "661",
34     "confidence": 0.98,
35     "camera_location": "Platform_2",
36     "head_count": 0
37 },
38 {
39     "timestamp": "2025-12-02T23:34:10",
40     "camera_id": "CAM_02",
41     "event_type": "crowd",
42     "object_id": null,
43     "confidence": 0.8,
44     "camera_location": "Main_hall",
45     "head_count": 10
46 },
47 {
48     "timestamp": "2025-12-02T05:21:56",
49     "camera_id": "CAM_05",
50     "event_type": "crowd",
51     "object_id": null,
52     "confidence": 0.94,
53     "camera_location": "Platform_2",
54     "head_count": 20

```

```

55 },
56 {
57     "timestamp": "2025-12-03T07:08:44",
58     "camera_id": "CAM_08",
59     "event_type": "person_entry",
60     "object_id": "972",
61     "confidence": 0.82,
62     "camera_location": "Main_hall",
63     "head_count": 0
64 },
65 {
66     "timestamp": "2025-12-03T05:06:54",
67     "camera_id": "CAM_04",
68     "event_type": "person_exit",
69     "object_id": "467",
70     "confidence": 0.87,
71     "camera_location": "Gate_01",
72     "head_count": 0
73 },
74 {
75     "timestamp": "2025-12-03T17:06:25",
76     "camera_id": "CAM_08",
77     "event_type": "crowd",
78     "object_id": null,
79     "confidence": 0.85,
80     "camera_location": "Platform_2",
81     "head_count": 7
82 }
83 ]

```

The stored data schema typically resembles:

```

{
    "timestamp": "2025-12-01T18:43:21",
    "camera_id": "CAM_03",

```

```

    "event_type": "car_exit",
    "object_id": "V_102",
    "confidence": 0.92,
    "location": "Gate_03"
  }

```

Feature Discription:

- timestamp : The times when the CV event was detected
- camera_id : A unique id given to each camera
- event_type : CV event name (car entry/exit, person entry/exit, crowd)
- object_id : a unique id given to detection
- confidence : confidence of the detection
- camera location : the location where camera is pointed to
- head_count : In the the crowd event number person detected

This structured representation ensures that downstream query interpretation by the LLM is both deterministic and context-aware.

3.4 LLM Integration for Surveillance Analytics

A Large Language Model (LLM) layer forms the reasoning engine of the system. The LLM performs two types of analyses:

3.4.1 Real-time Query Interpretation

Human users can directly ask questions such as:

- *“Which gate had the highest vehicle exit frequency yesterday?”*
- *“Show me locations where people gathered more than 50 individuals.”*
- *“Identify the camera with the most unattended object events this week.”*

The LLM translates these natural-language questions into:

1. Structured database queries, OR
2. Filter/aggregation logic applied to the event log dataset

3.4.2 Insight Generation

Beyond querying, the LLM generates synthesized insights such as:

- Temporal patterns (peak hours, anomaly clusters)
- Spatial correlations across multiple cameras
- Summary reports of high-risk or repetitive events

This enables intelligent forensic investigation without manually scanning massive surveillance datasets.

3.5 Insight Delivery to End Users

The final stage of the pipeline is the delivery of **Required / Asked Insights** back to the user. These insights can be delivered in multiple formats:

- Textual summaries
- Tabular reports
- Data visualizations
- Event timelines

By bridging CV perception with LLM reasoning, the system transforms raw surveillance footage into actionable security intelligence.

3.6 Implementation Summary

The implemented system:

- Integrates multiple CCTV nodes running optimized CV models.
- Stores structured event logs in a centralized database for unified access.
- Uses an LLM to process queries, generate insights, and perform temporal-spatial reasoning.
- Delivers meaningful forensic insights from both real-time and historical data.

This architecture demonstrates a scalable, intelligent, and efficient approach for next-generation surveillance analytics.

3.6.1 Demo

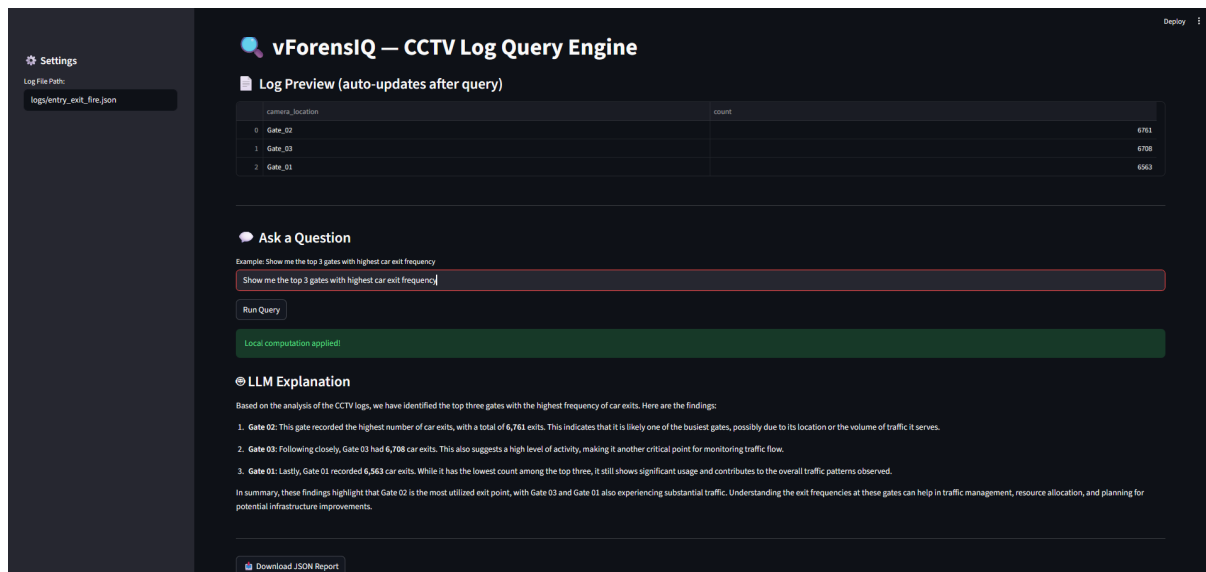


Figure 3.2: Demo screenshot with a sample query

The Figure 3.2 is a screenshot of streamlit based interactive dashboard designed for intelligent CCTV log analysis. It allows user to get insights from CCTV logs with natural language query.

Chapter 4

Conclusion

This project demonstrates the feasibility and effectiveness of integrating Large Language Models (LLMs) with structured surveillance event logs to enable intelligent forensic analysis. While computer vision (CV) models serve as the perception layer for detecting events and generating structured metadata, the primary contribution of this work lies in the reasoning capabilities provided by the LLM. By transforming raw CCTV footage into structured JSON logs, the system allows the LLM to interpret, summarize, filter and correlate events using natural language queries.

Through this integration, the system is capable of answering complex questions about historical and real-time activities in a surveillance environment—tasks that would traditionally require extensive manual review. The LLM-based analytic layer provides context-aware insights, temporal reasoning, and semantic interpretation of the event database, thereby significantly improving the usability of surveillance data for decision-making, investigation, and operational monitoring.

The results of this work confirm that LLMs can operate effectively over structured surveillance logs, mapping user intent to actionable database operations and generating human-friendly explanations. This positions LLMs not merely as conversational agents, but as powerful analytical engines capable of high-level surveillance intelligence. As such, CV is not the core focus of the system; rather, it acts as an enabling mechanism that supplies the LLM with reliable event-level information. The true value of the system emerges from the LLM's ability to reason over these structured logs and produce meaningful insights from vast surveillance datasets.

Chapter 5

Future Work

While the proposed CV-LLM architecture is functional and demonstrates strong potential, several opportunities exist for extending and optimizing the system, particularly on the LLM reasoning side. Future research directions include:

- **Fine-tuned LLMs for Surveillance Reasoning:** Current LLMs reason over event logs in a general-purpose manner. A fine-tuned or domain-adapted model could significantly improve event correlation, anomaly explanation, and temporal inference.
- **Advanced Query Translation and SQL Generation:** Future iterations can incorporate specialized query-planning modules or Text-to-SQL models to translate complex user queries into optimized database operations, improving accuracy and reducing latency.
- **Multi-LLM Collaboration or Agentic Pipelines:** A decentralized multi-agent architecture could divide tasks such as query understanding, data retrieval, correlation analysis, and summarization—creating a more robust analytic pipeline.
- **Enhanced Log Semantics and Event Ontologies:** Introducing an ontology-aware logging schema (e.g., temporal relationships, event hierarchy, causal links) would enable the LLM to perform more sophisticated reasoning, such as event prediction and root-cause analysis.
- **Human-in-the-Loop Feedback for Continual Learning:** User feedback on incorrect or incomplete insights can be used to refine the reasoning process of the

LLM over time, enabling adaptive improvement.

- **Lightweight and Edge-Optimized LLMs:** To reduce dependency on cloud infrastructure, future work may explore compressed, distilled, or quantized LLM variants that can operate in low-latency edge environments.
- **Contextual Multi-Modal Reasoning:** Although CV is not the focus, future systems could optionally allow LLMs to reference visual embeddings or keyframes for deeper reasoning, while still keeping event logs as the primary analytic source.

Overall, the future direction of this project lies in enhancing the LLM's analytical, reasoning, and retrieval capabilities rather than expanding the CV component. As LLMs continue to evolve, their ability to interpret structured surveillance logs, generate sophisticated insights, and assist in forensic investigation will grow considerably. The long-term goal is an autonomous, intelligent, and self-improving surveillance knowledge system—one where CV merely captures events, but the LLM serves as the core decision-making and reasoning engine.

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